Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(Artificial Intelligence & Machine Learning)

| | | | |
|---|---|---|---|
| **Academic Year:** | 2026-27 | **Name of Student:** | |
| **Semester:** | V | **Student ID:** | |
| **Class / Branch:** | TE – CSE (AI&ML) | **Roll No:** | |
| **Subject:** | WMD Lab | **Date of Submission:** | |

## Experiment No. 01

**Lab Outcome:** Demonstrate understanding of modern web and mobile development environments and tools.

**Aim**: To develop web page using HTML5.0 & CSS3 having table, form and lists with styling with properties and styles for Student Information System ***(Course Gap)***

**Software used**: HTML5/CSS3

**Theory: -**

### 1. Introduction to HTML5 and CSS3

HTML5 is the latest standard markup language used to structure content on the web. It introduced semantic elements such as <header>, <nav>, <main>, <section>, and <footer> that describe the meaning of content rather than just its appearance, improving accessibility, SEO, and code readability compared to older "div soup" markup. CSS3 is the styling language used to control the presentation of that content — layout, colour, typography, spacing, and responsiveness — and introduced powerful modern features such as custom properties, Flexbox, Grid, and media queries that this experiment relies on.

### 2. The CSS Box Model and box-sizing

Every HTML element is treated by the browser as a rectangular box made up of content, padding, border, and margin. By default (content-box), width and height apply only to the content area, so adding padding or a border increases the element's total rendered size — often causing layouts to overflow unexpectedly.

Setting box-sizing: border-box changes this behaviour so that padding and border are included inside the declared width and height, keeping element sizing predictable. Applying this through a universal reset ensures consistent sizing across all elements on the page:

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}
```

## 3. CSS Custom Properties (:root Variables)

The :root pseudo-class represents the document's root element and is commonly used to declare CSS custom properties (variables) such as colours, spacing, and radii in one place, for example --primary or --border-color. Because these variables are referenced throughout the stylesheet using var(--name), changing a single declaration instantly updates every element that uses it — similar to how a constant works in a programming language. This makes theming (including features like dark mode) and brand-colour changes trivial and consistent.

## 4. Layout Techniques: CSS Grid and Flexbox

Modern CSS provides two complementary layout engines:

| Layout Type | Tool | Best suited for |
|---|---|---|
| Macro (page-level) | CSS Grid | Two-dimensional layouts — arranging content into rows and columns simultaneously, e.g. splitting a page into a main content area and a sidebar using grid-template-columns: 2fr 1fr;. |
| Micro (component-level) | CSS Flexbox | One-dimensional alignment — distributing or aligning items along a single row or column, e.g. display: flex; justify-content: space-between; to push a logo and navigation links to opposite ends of a header. |

A useful rule of thumb: use Grid when both rows and columns matter to the page structure, and use Flexbox when only one direction (a row or a column) needs to be aligned or distributed, such as inside a navigation bar or a card.

## 5. Positioning: Sticky Elements and Stacking Context

Setting position: sticky; top: 0; on an element (such as a page header) keeps it fixed at the top of the viewport once the page scrolls past its normal position, while still behaving like a normal element before that point. For sticky positioning to work, the element needs a threshold property (top, bottom, left, or right) and none of its parent containers may have overflow set to hidden, scroll, or auto.

The z-index property controls the stacking order of overlapping elements along the page's z-axis: elements with a higher z-index render above elements with a lower one. This is important for a sticky header, which needs a high z-index (e.g. 100) so that content scrolling underneath it — such as a table or form — does not visually bleed over the top.

## 6. Forms and Accessibility (a11y)

Accessible, semantic form markup improves usability for all users, including those relying on assistive technology such as screen readers:

- <fieldset> and <legend>: group related form controls logically (e.g. term details vs. contact details) and give the group a text label that assistive technology announces to the user.
- <label for="id">: programmatically associates a label with its input by matching the label's for attribute to the input's id, which expands the clickable area and lets screen readers announce the field's purpose.
- HTML5 constraint validation: attributes such as required, type="email", and type="date" let the browser validate input natively, without any JavaScript.

Two useful pseudo-classes give this validation visual feedback:

```
input:focus {
  border-color: var(--primary);
  box-shadow: 0 0 0 3px var(--primary-light);
}

input:focus:invalid {
  border-color: var(--danger);
  box-shadow: 0 0 0 3px var(--danger-light);
}
```

:focus highlights the field currently receiving keyboard input, and :invalid flags a field whose value fails HTML5 validation — combined as :focus:invalid, the input's border turns red only while the user is actively editing an invalid value.

## 7. Tables and Semantic Lists for Data Display

The <table> element remains the only semantically correct way to present tabular data — rows and columns of related values — and should not be used for general page layout. When a table holds many rows, wrapping it in a container with a fixed max-height and overflow-y: auto creates a local scrollbar, while applying position: sticky; top: 0; to the <th> elements keeps the column headers visible as the body scrolls, so context is never lost. border-collapse: collapse also removes the doubled borders that otherwise appear between adjacent cells.

Semantic elements can also be creatively restyled: an unordered list (<ul>) with list-style: none and a coloured border-left can be turned into a visual timeline, using modifier classes (e.g. .academic, .warning, .success) to vary the accent colour per item without duplicating the base styles — an example of the Block-Element-Modifier (BEM) approach to CSS class naming.

## 8. Responsive Web Design and Media Queries

Responsive design ensures a page adapts cleanly to different screen sizes rather than breaking or requiring horizontal scrolling. CSS media queries apply a block of styles only when the viewport matches a given condition, such as a maximum width:

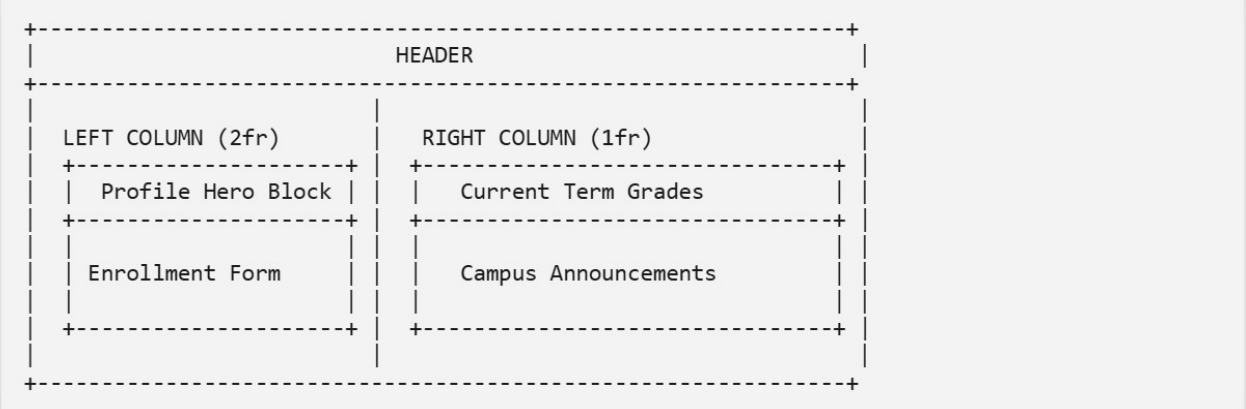
```
@media (max-width: 1024px) {
  main {
    grid-template-columns: 1fr; /* collapse two columns into one */
  }
}
```

A common strategy is desktop-first degradation: the page is designed for large screens first, and media queries at progressively smaller breakpoints (e.g. 1024px for tablets, 640px for phones) simplify the layout — collapsing multi-column grids into a single column, and letting flexible components such as Flexbox rows wrap onto new lines — so the interface remains usable on any device.

# Implementation

## Design Overview

The page is built as a Student Information System (SIS) dashboard using an asymmetric desktop layout: a two-column CSS Grid (main content on the left, secondary widgets on the right) that collapses into a single stacked column on smaller screens. The overall structure is outlined below:




### Code Walkthrough

### 1. Global Theming and Box Reset

Design tokens are declared once in :root and reused everywhere via var(), and a universal box-sizing reset keeps element sizing predictable:

```
:root {
  --primary: #2563eb;
  --success: #16a34a;
  --danger: #dc2626;
  --warning: #ca8a04;
  --text-main: #1e293b;
  --border-color: #e2e8f0;
  --radius: 0.75rem;
}

* { box-sizing: border-box; margin: 0; padding: 0; }
```

## 2. Sticky Header and Macro/Micro Layout

The <header> uses Flexbox to push the logo and nav to opposite ends, and stays docked at the top of the viewport while scrolling:

```
header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  position: sticky;
  top: 0;
  z-index: 100;
}
```

The <main> dashboard area uses CSS Grid for its two-column macro layout:

```
main {
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 2rem;
}
```

## 3. Accessible Enrollment Form

Related inputs are grouped with <fieldset>/<legend>, and native HTML5 validation is used instead of JavaScript:

```
<fieldset>
  <legend>Academic Term Enrollment</legend>
  <div class="form-group">
    <label for="semester">Target Semester *</label>
    <select id="semester" required>...</select>
  </div>
</fieldset>
```

## 4. Data Table and Announcements Timeline

The grades table is wrapped in a scrollable container with a sticky header row:

```
.table-container {
  max-height: 350px;
  overflow-y: auto;
}
th {
  position: sticky;
  top: 0;
}
```

The announcements list is restyled from a plain <ul> into a timeline using a coloured left border and modifier classes:

```
.timeline-item {
  border-left: 3px solid var(--border-color);
  padding-left: 1.5rem;
}
.timeline-item.warning { border-left-color: var(--warning); }
.timeline-item.success { border-left-color: var(--success); }
```

### 5. Responsive Breakpoints

Two media query breakpoints progressively simplify the layout for tablets and phones:

```
@media (max-width: 1024px) {
  main { grid-template-columns: 1fr; }
}
@media (max-width: 640px) {
  .form-grid { grid-template-columns: 1fr; }
  .profile-hero { flex-direction: column; text-align: center; }
}
```

## Source Code

### Student Information System.html

D: > AY_2026-27 > ODD > WMD_VJshNair > Lab > Exp-1 > Student Information System.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Information System Dashboard</title>
    <style>
        /* --- CSS VARIABLES & THEME SETUP --- */
        :root {
            --bg-color: #f4f6f9;
            --card-bg: #ffffff;
            --primary: #2563eb;
            --primary-light: #dbeafe;
            --success: #16a34a;
            --success-light: #dcfce7;
            --danger: #dc2626;
            --danger-light: #fee2e2;
            --warning: #ca8a04;
            --warning-light: #fef9c3;
            --text-main: #1e293b;
            --text-muted: #64748b;
            --border-color: #e2e8f0;
            --radius: 0.75rem;
            --shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05), 0 2px 4px -1px rgba(0, 0, 0, 0.03);
        }

        /* --- BASE STYLES --- */
        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
        }

        body {
            font-family: 'Segoe UI', system-ui, -apple-system, sans-serif;
            background-color: var(--bg-color);
            color: var(--text-main);
            line-height: 1.5;
            padding-bottom: 3rem;
        }

        /* --- HEADER NAVBAR --- */
        header {
            background-color: var(--card-bg);
            border-bottom: 1px solid var(--border-color);
            padding: 1rem 2rem;
            display: flex;
            justify-content: space-between;
            align-items: center;
            position: sticky;
            top: 0;
            z-index: 100;
        }
```

```css
.logo h1 {
    font-size: 1.25rem;
    color: var(--primary);
    font-weight: 700;
}

.user-nav {
    display: flex;
    align-items: center;
    gap: 1.5rem;
}

.notif-badge {
    background-color: var(--danger);
    color: white;
    padding: 0.15rem 0.4rem;
    border-radius: 20px;
    font-size: 0.75rem;
    font-weight: bold;
}

/* --- DASHBOARD MESH LAYOUT --- */
main {
    max-width: 1400px;
    margin: 2rem auto;
    padding: 0 1.5rem;
    display: grid;
    grid-template-columns: 2fr 1fr;
    gap: 2rem;
}

section {
    background-color: var(--card-bg);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 1.75rem;
    margin-bottom: 2rem;
    border: 1px solid var(--border-color);
}

.left-column, .right-column {
    display: flex;
    flex-direction: column;
}

/* --- COMPONENT 1: PROFILE HERO --- */
.profile-hero {
    display: flex;
    flex-wrap: wrap;
    align-items: center;
    gap: 2rem;
}

.profile-avatar {
    width: 90px;
    height: 90px;
    background: linear-gradient(135deg, var(--primary), #3b82f6);
    border-radius: 50%;
    display: flex;
    align-items: center;
```

```
        justify-content: center;
        color: white;
        font-size: 2rem;
        font-weight: bold;
    }

    .profile-meta {
        flex: 1;
        min-width: 200px;
    }

    .profile-meta h2 {
        font-size: 1.5rem;
        margin-bottom: 0.25rem;
    }

    .profile-meta p {
        color: var(--text-muted);
        font-size: 0.9rem;
    }

    .profile-stats {
        display: flex;
        gap: 1rem;
    }

    .stat-block {
        background-color: var(--bg-color);
        padding: 0.75rem 1.2rem;
        border-radius: var(--radius);
        text-align: center;
    }

    .stat-val {
        display: block;
        font-size: 1.25rem;
        font-weight: 700;
        color: var(--primary);
    }

    .stat-label {
        font-size: 0.75rem;
        color: var(--text-muted);
        text-transform: uppercase;
        letter-spacing: 0.05em;
    }

    /* --- COMPONENT 2: ENROLLMENT FORM --- */
    fieldset {
        border: none;
        margin-bottom: 1.5rem;
    }

    legend {
        font-size: 1.1rem;
        font-weight: 600;
        margin-bottom: 1rem;
        color: var(--primary);
        width: 100%;
        border-bottom: 1px solid var(--border-color);
        padding-bottom: 0.25rem;
```

```
176         }
177
178         .form-grid {
179             display: grid;
180             grid-template-columns: repeat(2, 1fr);
181             gap: 1.25rem;
182         }
183
184         .form-group {
185             display: flex;
186             flex-direction: column;
187             gap: 0.35rem;
188         }
189
190         .form-group.full-width {
191             grid-column: span 2;
192         }
193
194         label {
195             font-size: 0.85rem;
196             font-weight: 600;
197             color: var(--text-main);
198         }
199
200         input, select {
201             width: 100%;
202             padding: 0.65rem 0.8rem;
203             border: 1px solid var(--border-color);
204             border-radius: 6px;
205             font-family: inherit;
206             font-size: 0.95rem;
207             color: var(--text-main);
208             background-color: #fff;
209             transition: border-color 0.2s ease, box-shadow 0.2s ease;
210         }
211
212         /* Interactive Focus & Validation CSS states */
213         input:focus, select:focus {
214             outline: none;
215             border-color: var(--primary);
216             box-shadow: 0 0 0 3px var(--primary-light);
217         }
218
219         input:focus:invalid {
220             border-color: var(--danger);
221             box-shadow: 0 0 0 3px var(--danger-light);
222         }
223
224         .btn-submit {
225             background-color: var(--primary);
226             color: white;
227             border: none;
228             padding: 0.75rem 1.5rem;
229             font-size: 1rem;
230             font-weight: 600;
231             border-radius: 6px;
232             cursor: pointer;
233             transition: background-color 0.2s;
234             align-self: flex-start;
235         }
236
```

```
237         .btn-submit:hover {
238             background-color: #1d4ed8;
239         }
240
241         /* --- COMPONENT 3: GRADES TABLE --- */
242         .table-container {
243             overflow-x: auto;
244             max-height: 350px; /* Sticky container constraints */
245             overflow-y: auto;
246             border: 1px solid var(--border-color);
247             border-radius: 6px;
248         }
249
250         table {
251             width: 100%;
252             border-collapse: collapse;
253             text-align: left;
254             font-size: 0.9rem;
255         }
256
257         th, td {
258             padding: 0.85rem 1rem;
259             border-bottom: 1px solid var(--border-color);
260         }
261
262         th {
263             background-color: #f8fafc;
264             color: var(--text-main);
265             font-weight: 600;
266             position: sticky;
267             top: 0;
268             z-index: 10;
269         }
270
271         tr:nth-child(even) td {
272             background-color: #f8fafc;
273         }
274
275         tr:hover td {
276             background-color: #f1f5f9;
277         }
278
279         /* Pill Badges */
280         .badge {
281             display: inline-block;
282             padding: 0.25rem 0.6rem;
283             border-radius: 50px;
284             font-size: 0.75rem;
285             font-weight: 600;
286             text-transform: uppercase;
287         }
288
289         .badge.pass { background-color: var(--success-light); color: var(--success); }
290         .badge.fail { background-color: var(--danger-light); color: var(--danger); }
291         .badge.progress { background-color: var(--warning-light); color: var(--warning); }
292
293         /* --- COMPONENT 4: TIMELINE LISTS --- */
294         .timeline {
295             list-style: none;
296             position: relative;
297         }
298
```

```
            .timeline-item {
                position: relative;
                padding-left: 1.5rem;
                margin-bottom: 1.5rem;
                border-left: 3px solid var(--border-color);
            }

            /* Target Left Accent Styles via explicit border color classes */
            .timeline-item.academic { border-left-color: var(--primary); }
            .timeline-item.warning { border-left-color: var(--warning); }
            .timeline-item.success { border-left-color: var(--success); }

            .timeline-item:last-child {
                margin-bottom: 0;
            }

            .timeline-time {
                font-size: 0.75rem;
                color: var(--text-muted);
                font-weight: 600;
                display: block;
                margin-bottom: 0.15rem;
            }

            .timeline-title {
                font-size: 0.95rem;
                font-weight: 600;
                margin-bottom: 0.25rem;
            }

            .timeline-desc {
                font-size: 0.85rem;
                color: var(--text-muted);
            }

            /* --- RESPONSIVE MEDIA BREAKPOINTS --- */
            @media (max-width: 1024px) {
                main {
                    grid-template-columns: 1fr;
                }
            }

            @media (max-width: 640px) {
                .form-grid {
                    grid-template-columns: 1fr;
                }
                .form-group.full-width {
                    grid-column: span 1;
                }
                .profile-hero {
                    flex-direction: column;
                    text-align: center;
                }
                .profile-stats {
                    width: 100%;
                    justify-content: center;
                }
            }
        </style>
    </head>
    <body>
```

```
    <header>
        <div class="logo">
            <h1>A. P. Shah Institute of Technology  || <span style="font-weight: 400; color: var(--text-muted);">Student Information System</span></h1
        </div>
        <nav class="user-nav">
            <div>Alerts <span class="notif-badge">3</span></div>
            <strong style="font-size: 0.9rem;">Admin View</strong>
        </nav>
    </header>

    <main>

        <div class="left-column">

            <section class="profile-hero">
                <div class="profile-avatar">VN</div>
                <div class="profile-meta">
                    <h2>Vijesh Nair</h2>
                    <p><strong>Major:</strong> B.E. Computer Science & Engineering ( AI & Ml)</p>
                    <p><strong>Student ID:</strong> #2026-98431</p>
                </div>
                <div class="profile-stats">
                    <div class="stat-block">
                        <span class="stat-val">8.84</span>
                        <span class="stat-label">CGPA</span>
                    </div>
                    <div class="stat-block">
                        <span class="stat-val">92%</span>
                        <span class="stat-label">Attendance</span>
                    </div>
                </div>
            </section>

            <section>
                <form action="#" method="POST" onsubmit="event.preventDefault();">

                    <fieldset>
                        <legend>Academic Term Enrollment</legend>
                        <div class="form-grid">
                            <div class="form-group">
                                <label for="semester">Target Semester *</label>
                                <select id="semester" required>
                                    <option value="">Select Option</option>
                                    <option value="fall2026">Odd 2026</option>
                                    <option value="spring2027">Even 2027</option>
                                </select>
                            </div>
                            <div class="form-group">
                                <label for="enroll-date">Registration Date *</label>
                                <input type="date" id="enroll-date" required>
                            </div>
                        </div>
                    </fieldset>

                    <fieldset>
                        <legend>Course Preferences & Contact</legend>
                        <div class="form-grid">
                            <div class="form-group">
                                <label for="primary-elective">Primary Elective Focus *</label>
                                <select id="primary-elective" required>
                                    <option value="">Choose Elective Path</option>
                                    <option value="ai">Agentic AI</option>
                                    <option value="cyber">Cybersecurity Infrastructure</option>
```

```
                                <option value="cloud">Cloud Architecture</option>
                            </select>
                        </div>
                        <div class="form-group">
                            <label for="student-email">Academic Email *</label>
                            <input type="email" id="student-email" placeholder="vijesh.nair@apsit.edu.in" required>
                        </div>
                        <div class="form-group full-width">
                            <label for="notes">State your Special Interest</label>
                            <input type="text" id="notes" placeholder="Specify any scheduling achievements or career pursue plans etc.,">
                        </div>
                    </div>
                </fieldset>

                <button type="submit" class="btn-submit">Confirm Registration Selection</button>
            </form>
        </section>
    </div>

    <div class="right-column">

        <section>
            <h3 style="margin-bottom: 1rem; font-size: 1.1rem;">Current Term Grades</h3>
            <div class="table-container">
                <table>
                    <thead>
                        <tr>
                            <th>Code</th>
                            <th>Grade</th>
                            <th>Status</th>
                        </tr>
                    </thead>
                    <tbody>
                        <tr>
                            <td><strong>CS-301</strong><br><span style="font-size:0.75rem; color:var(--text-muted)">Data Structures</span></td>
                            <td>A</td>
                            <td><span class="badge pass">Pass</span></td>
                        </tr>
                        <tr>
                            <td><strong>CS-304</strong><br><span style="font-size:0.75rem; color:var(--text-muted)">Database Sys</span></td>
                            <td>B+</td>
                            <td><span class="badge pass">Pass</span></td>
                        </tr>
                        <tr>
                            <td><strong>MATH-210</strong><br><span style="font-size:0.75rem; color:var(--text-muted)">Linear Algebra</span></td>
                            <td>--</td>
                            <td><span class="badge progress">In Progress</span></td>
                        </tr>
                        <tr>
                            <td><strong>EEE-102</strong><br><span style="font-size:0.75rem; color:var(--text-muted)">Basic Electricals</span></td>
                            <td>E</td>
                            <td><span class="badge fail">Fail</span></td>
                        </tr>
                    </tbody>
                </table>
            </div>
        </section>

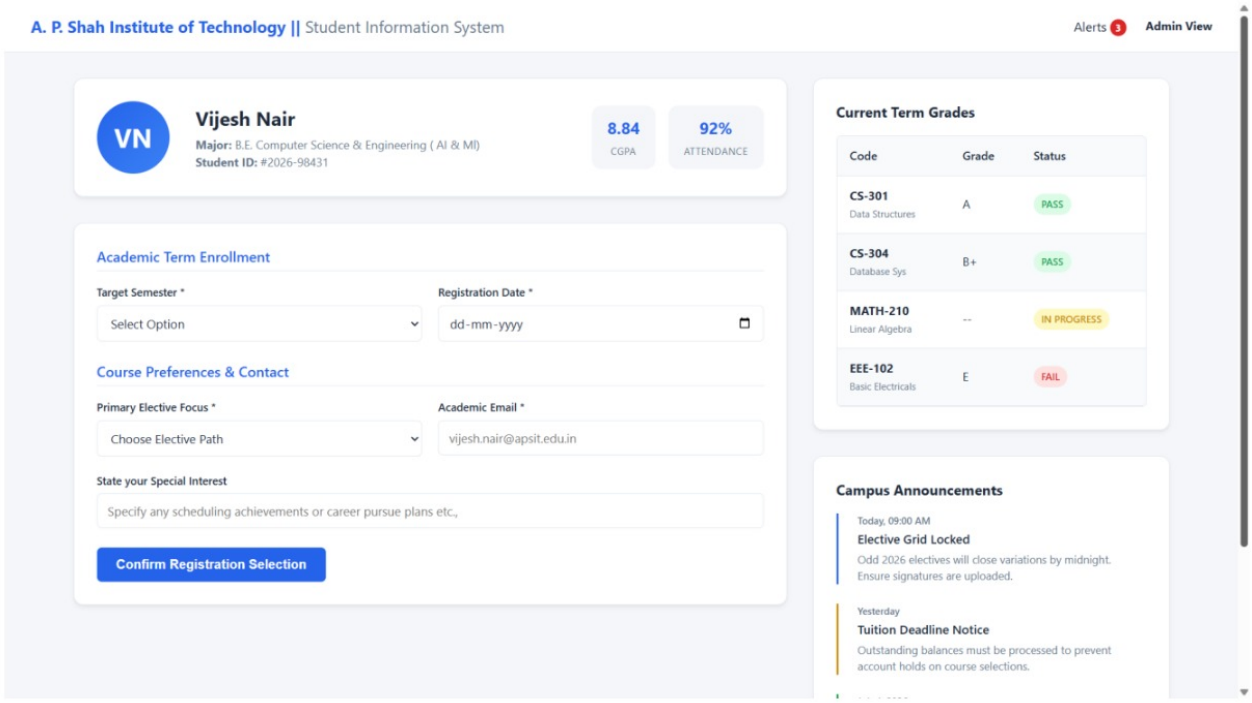
        <section>
            <h3 style="margin-bottom: 1rem; font-size: 1.1rem;">Campus Announcements</h3>
```

```
                <ul class="timeline">
                    <li class="timeline-item academic">
                        <span class="timeline-time">Today, 09:00 AM</span>
                        <h4 class="timeline-title">Elective Grid Locked</h4>
                        <p class="timeline-desc">Odd 2026 electives will close variations by midnight. Ensure signatures are uploaded.</p>
                    </li>
                    <li class="timeline-item warning">
                        <span class="timeline-time">Yesterday</span>
                        <h4 class="timeline-title">Tuition Deadline Notice</h4>
                        <p class="timeline-desc">Outstanding balances must be processed to prevent account holds on course selections.</p>
                    </li>
                    <li class="timeline-item success">
                        <span class="timeline-time">July 1, 2026</span>
                        <h4 class="timeline-title">Hackathon Registration Live</h4>
                        <p class="timeline-desc">Annual multi-disciplinary system building event opens applications today.</p>
                    </li>
                </ul>
            </section>
        </div>

    </main>

</body>
</html>
```

## Output:




## Conclusion: -

Thus, we have illustrated the development of web page using HTML5.0 & CSS3 having table, form and lists with styling with properties and styles for Student Information System.